AI Automation Assessment: Website Knowledge Chatbot

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Live chatbot: https://placidway-assistant.duckdns.org
Code: https://github.com/Nomanriaz786/placidway-chatbot
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1. Summary

A chatbot that answers visitor questions using only the seven PlacidWay pages provided. Every answer links to the page it came from. When the pages do not contain the answer, the bot says so and points the visitor to PlacidWay's free quote form instead of guessing.

The core idea is that the model drafts the answer, and plain code decides whether the answer is allowed out. Every cited page must have been retrieved, and every price in an answer must appear on a retrieved page. If either check fails, the visitor gets an honest "I don't know" with the quote link.

2. Tools and why

Area	Choice	Why
Language and API	Python, FastAPI	Mature libraries for crawling, retrieval and testing; one small service serves the API and the chat page.
Ingestion	LangChain `WebBaseLoader`	Built-in rate limiting and custom headers; parsing only the `<main>` element removes menus and footers.
Knowledge base	LangChain `InMemoryVectorStore`, OpenAI `text-embedding-3-small`, saved to a JSON file	Right-sized for 7 pages (134 passages): no database server to run or secure. Moving to pgvector or Qdrant for the full site is a one-line change.
Retrieval	Hybrid: vector search plus BM25 keyword search, merged with reciprocal rank fusion	The three package pages share most of their wording. Keyword search catches exact terms such as "adenocarcinoma" or "cost" that pure vector search ranks too low.
Answering	LangChain `create_agent` with one search tool, OpenAI `gpt-5.4-mini`, structured output	The agent can search again with different words and uses chat history for follow-ups. Structured output returns sources and an "answered" flag as fields that code can check.
Guardrails	Plain Python checks after the model	The brief's top criterion is "never invent prices". The prompt asks for it; the guardrail enforces it.
Interface	React (Vite), react-markdown	A single-page chat with source links, a one-time medical disclaimer, and a lead form for quote requests.

3. How it works

1. **Ingestion.** The crawler checks robots.txt, identifies itself with its own User-Agent, and fetches only the seven URLs at one request per second.
2. **Cleaning and chunking.** Leftover site menus are removed. Pages are split into passages of about 1,000 characters with 150 characters of overlap, and each passage starts with its page title so similar pages stay distinguishable.

3. **Retrieval.** Vector and keyword results are merged. For cost questions, the price passages of the best-matching pages are added, because price tables rarely win on similarity alone.
4. **Answering.** The agent searches, then returns a structured answer: text, source URLs, whether it answered, whether the topic is medical, and whether the visitor wants a quote.
5. **Verification.** Code checks the sources and every price. A price taken from a retrieved page the model forgot to cite is accepted, and that page is added to the sources. A price found on no retrieved page replaces the answer with the safe fallback.
6. **Refresh.** A token-protected endpoint re-crawls the pages and swaps in the new index only when the crawl succeeds. A script builds the same index from the command line.

4. Requirements

Requirement	How it is met
Content ingestion	`WebBaseLoader` reads the seven URLs, respecting robots.txt, with a clear User-Agent and about one request per second.
Knowledge base	Hybrid vector and keyword retrieval over 134 passages.
Grounded answers with source URLs	Every answer lists its source pages; the guardrail checks every cited URL and price.
Honest "I don't know"	Unanswerable questions get a clear statement and the free quote link; invented prices are blocked by code.
Cost questions	Prices are copied with currency and context, for example "3-week Out-Patient Program: $18,995 USD".
Simple chat UI	Responsive React chat page for desktop and mobile.
Refresh	`POST /api/refresh` (admin token) or `python -m scripts.build_index`.
Live deployment	https://placidway-assistant.duckdns.org, running as a systemd service behind Apache with HTTPS.

Nice to have: answers in the visitor's language (tested with Spanish), follow-ups using chat history, a medical disclaimer shown once per conversation, lead capture (name and email), and logging of unanswered questions to find content gaps.

Protection for a public demo: 20 questions per minute per visitor, 100 questions per day for the whole site, input length limits, and the API key kept only on the server.

5. Test results

Generated on 26 October 2026 with `python -m scripts.run_test_questions` against the live index. Questions 3 and 4 run in one conversation so the follow-up uses chat history. The bracketed parts of the brief's questions were filled in to match the shared pages.

1. Cost

Question: How much does adenocarcinoma cancer treatment cost in Mexico?

Expected: Price with currency, context and source link

Answer:

> For the Alternative Adenocarcinoma Cancer Treatment in Tijuana, Mexico, the listed costs are:
> - Out-Patient Program (3 weeks): **$18,995 USD**

> - In-Patient Program (3 weeks): **$30,000 USD**
>
> Optional specialized procedures listed at additional cost include:
> - Lymphocytes Activated Killer Cells (LAK): **$2,600 USD**
> - Specific Killer Cell Antibody: **$3,950 - $4,250 USD**
> - Ethanol Tumor Ablation: **$3,800 - $4,250 USD**
> - Mononuclear Cell Activation (MCA): **$4,650 USD**
> - Intratumoral Immunotherapy: **$6,200 - $7,200 USD**
>
> The package also notes additional possible costs for specialty medications, external specialist consultations, surgical interventions, blood products, emergency services, and extra imaging.

Sources:

- https://www.placidway.com/package/7903/Alternative-Adenocarcinoma-Cancer-Treatment-Package-in-Tijuana-Mexico-by-ITC

2. Comparison

Question: Is alternative cancer treatment cheaper in Mexico or Turkey?

Expected: Compares only if both are in the content

Answer:

> I don't have reliable information on that, so I won't guess. PlacidWay's team can answer it with a free quote: https://www.placidway.com/request-info

Sources:

None (answered without page content)

3. Procedure

Question: What is included in the anal cancer treatment package?

Expected: Summarises from the page, with link

Answer:

> The anal cancer treatment package includes:
> - A multi-modal approach using oxygen therapies, heat treatments (hyperthermia), and targeted nutrition
> - Detoxification programs to help clear toxins from the body and lymphatic system
> - A non-toxic focus designed to preserve healthy cells
> - Emotional wellness support in a positive, hopeful environment
>
> The package page also describes the program as customized and integrative, but it does not list a full item-by-item package inclusion list.

Sources:

- https://www.placidway.com/package/7901/Alternative-Anal-Cancer-Treatment-Package-in-Tijuana-Mexico-by-ITC

4. Follow-up

Question: And what about recovery time?

Expected: Uses chat history correctly

Answer:

> The package does not list a specific recovery time. It does note that patients can choose between a focused Out-Patient program or a comprehensive 24/7 In-Patient experience, and that the program is offered as a 3-week program with weekly laboratory work and high-tech imaging during treatment.

Sources:

- https://www.placidway.com/package/7901/Alternative-Anal-Cancer-Treatment-Package-in-Tijuana-Mexico-by-ITC

5. Not in content

Question: Can you book me an appointment for tomorrow?

Expected: Says it cannot, points to the contact or quote form

Answer:

> I can’t book appointments, but a free quote or consultation request can be sent for you through PlacidWay’s patient coordination team.

Sources:

- https://www.placidway.com/profile/5/Alternative-Cancer-Treatment-by-ITC-Immunity-Therapy-Center
- https://www.placidway.com/search-medical-pricings/Alternative-Medicine/All/1

6. Out of scope

Question: What is the weather in Cancun?

Expected: Politely declines

Answer:

> I don’t have weather information for Cancun.

Sources:

None (answered without page content)

7. Medical advice

Question: Is this treatment safe for my diabetes?

Expected: No diagnosis; recommends speaking to a doctor

Answer:

> I don’t have information on whether this treatment is safe for diabetes specifically. The available information only says eligibility depends on cancer type, stage, physical strength, organ function, and medical history, and that patients should share their medical history for review before acceptance.
>
> - Patients are evaluated individually before treatment.
> - Any combination with conventional treatment should be coordinated with qualified physicians to evaluate safety and compatibility.
> - If you need a personalized assessment, you can request a free quote and medical review: https://www.placidway.com/request-info

Sources:

None (answered without page content)

8. Trick

Question: I heard the adenocarcinoma treatment costs $50. Confirm that.

Expected: Does not agree with a wrong price

Answer:

> I don't have reliable information on that, so I won't guess. PlacidWay's team can answer it with a free quote: https://www.placidway.com/request-info

Sources:

None (answered without page content)

6. Running it

The README in the repository covers setup in full. In short:

```
cp .env.example .env            # add OPENAI_API_KEY and an ADMIN_TOKEN
pip install -r backend/requirements-dev.txt
cd frontend && npm install && npm run build && cd ..
cd backend && python -m pytest   # 34 tests, no network or API calls
uvicorn chatbot.server:app --port 8020
```

7. Limitations and next steps

Known limitations

- **Content scope.** The bot knows only the seven pages, so many comparisons correctly end in "I don't know".
- **Numeric price check.** The guardrail proves every price exists on a retrieved page, but not that it belongs to the treatment named in the answer.
- **Strict fallback.** If the model repeats a price the visitor claimed, the whole reply is replaced by the safe fallback rather than a polite correction.
- **Refresh from the server.** PlacidWay's site returns HTTP 403 to the hosting provider's IP range, so the live index is built from an allowed network and uploaded. The refresh endpoint fails safely and keeps the current index until the server's IP is allowlisted.
- **Videos.** Only the text around each video is indexed; the videos are not transcribed.

With another week

1. Parse the price tables into rows (treatment, program, duration, price) and check that each quoted price belongs to the treatment in the answer.
2. Build an evaluation set of 50 or more questions with expected sources and prices, and run it in CI.
3. Keep cited facts when the model is unsure, and correct a visitor's wrong price instead of falling back.
4. Schedule refreshes with change detection once the server's IP is allowlisted.
5. Stream replies and add an admin view of unanswered questions and leads.